

DPO

Direct Preference Optimization

东川路第一可爱猫猫虫

主要内容

- RLHF 目标函数最优解的推导

$$\pi^*(y|x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right)$$

- DPO loss函数
- DPO 如何更新
- DPO 训练时的监控量

感谢

宽人克己是謂上策

阿萨李逍遥张小凡

蕾西亚万岁啊我透

bcd fgh

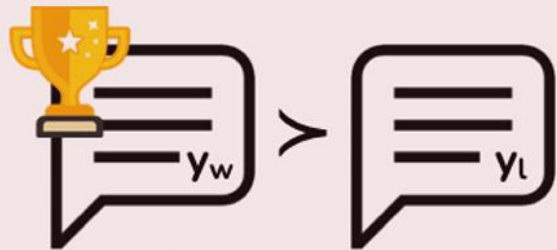
居居熊哒

JasonCZH4

RLHF (InstructGPT 2022.03)

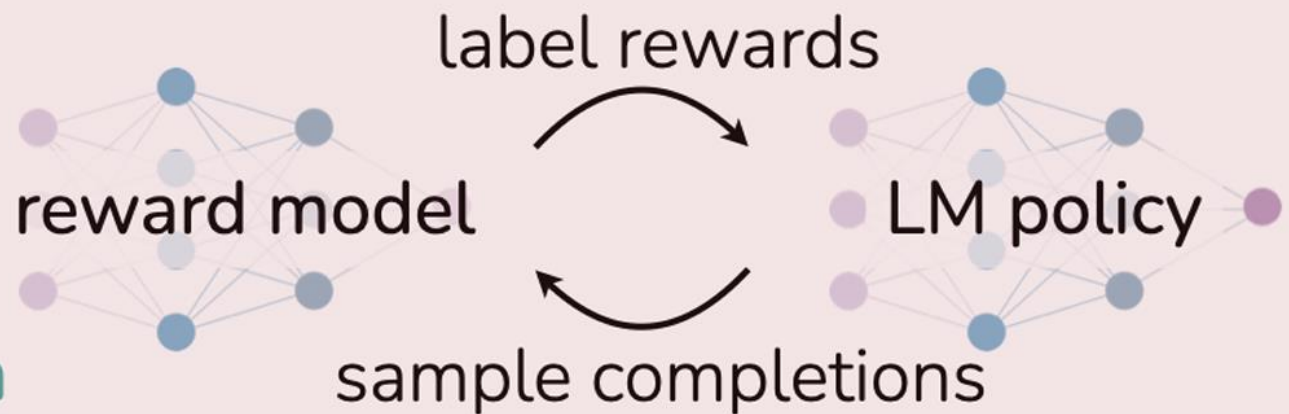
Reinforcement Learning from Human Feedback (RLHF)

x : "write me a poem about
the history of jazz"



preference data

maximum
likelihood

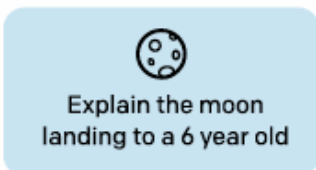


reinforcement learning

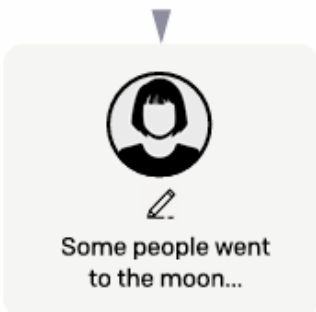
Step 1

Collect demonstration data, and train a supervised policy.

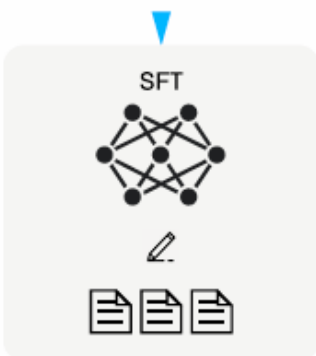
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



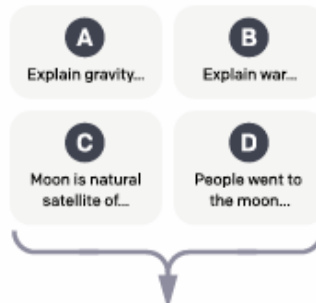
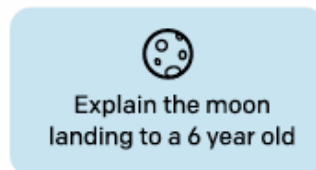
This data is used to fine-tune GPT-3 with supervised learning.



Step 2

Collect comparison data, and train a reward model.

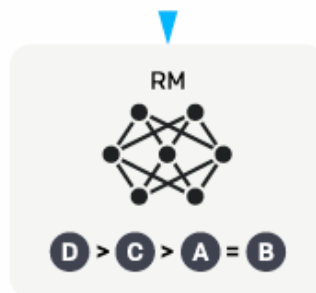
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



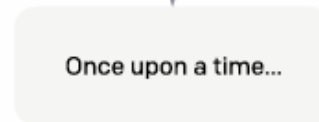
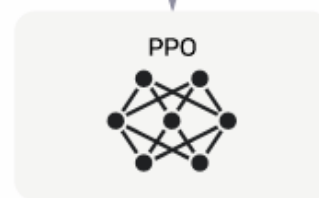
Step 3

Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.



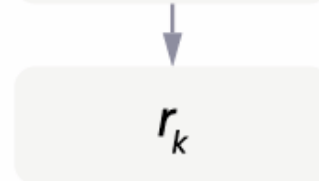
The policy generates an output.

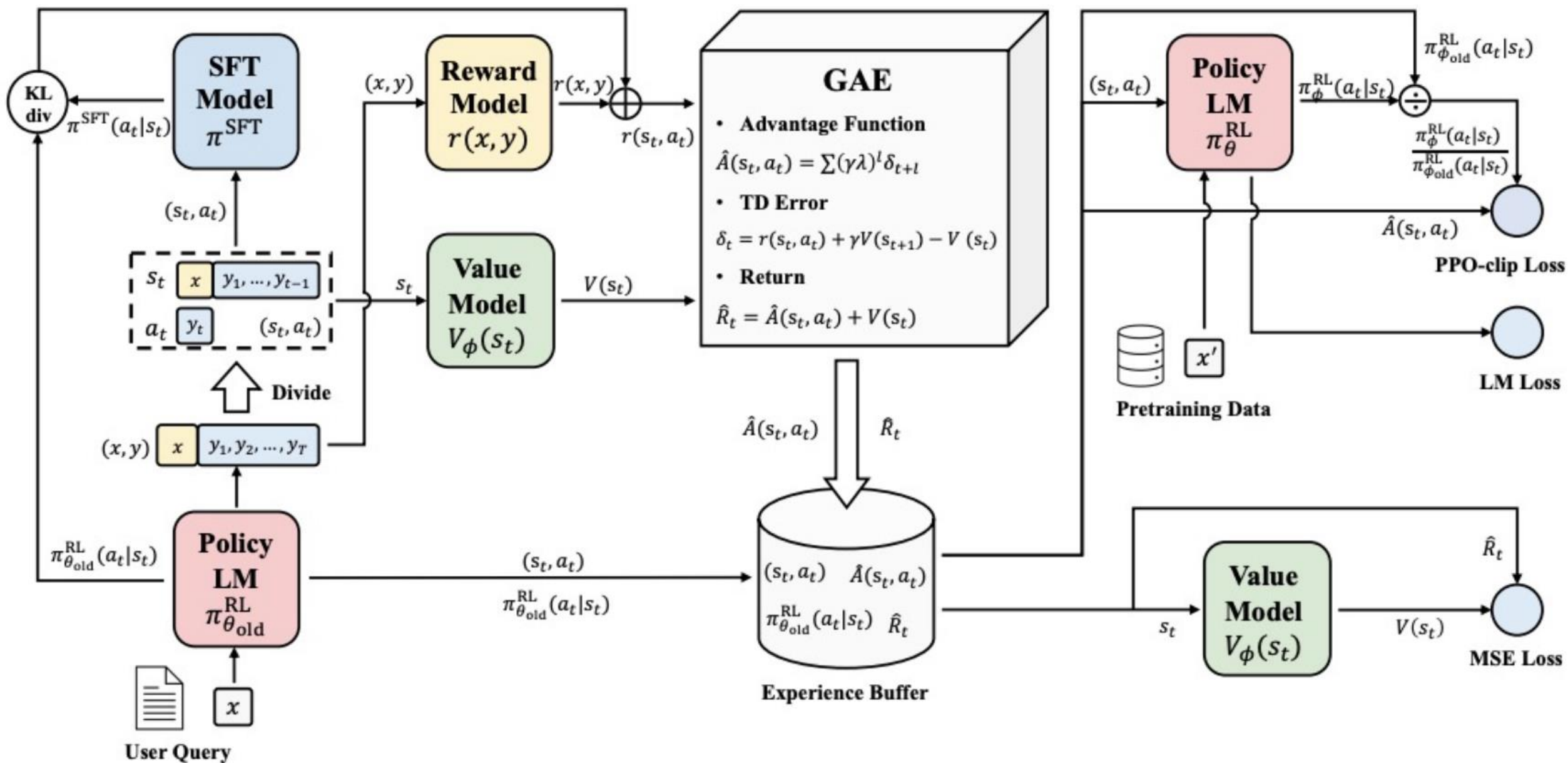


The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.





目标函数

- $\max_{\pi_{\theta}} E_{x \sim \mathbb{D}, y \sim \pi_{\theta}(y|x)} [r_{\phi}(x, y)]$
- $\max_{\pi_{\theta}} \underbrace{E_{x \sim \mathbb{D}, y \sim \pi_{\theta}(y|x)} [r_{\phi}(x, y)]}_{\text{第一部分：奖励}} - \underbrace{\beta E_{x \sim \mathbb{D}} D_{\text{KL}}[\pi_{\theta}(y|x) | \pi_{\text{ref}}(y|x)]}_{\text{第二部分：约束}}$
- DPO建立了从人类偏好到策略的映射
无需再训练奖励模型
- $\max_{\pi} E_{x \sim \mathbb{D}, y \sim \pi} [r(x, y)] - \beta D_{\text{KL}}[\pi(y|x) | \pi_{\text{ref}}(y|x)]$
是有最优解的
- 接下来我们将求出这个最优解

- KL散度: $D_{\text{KL}}(P|Q) = E_{y \sim P} \left[\log \frac{P(y)}{Q(y)} \right]$
- $D_{\text{KL}}[\pi(y|x)|\pi_{\text{ref}}(y|x)] = E_{y \sim \pi(y|x)} \left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} \right]$
- $\max_{\pi_{\theta}} E_{x \sim \mathbb{D}, y \sim \pi_{\theta}(y|x)} [r_{\phi}(x, y)] - \beta E_{x \sim \mathbb{D}} D_{\text{KL}}[\pi_{\theta}(y|x)|\pi_{\text{ref}}(y|x)]$
 $= E_{x \sim \mathbb{D}} [E_{y \sim \pi(y|x)} [r(x, y)] - \beta \cdot D_{\text{KL}}[\pi(y|x)|\pi_{\text{ref}}(y|x)]]$
 $= E_{x \sim \mathbb{D}} \left[E_{y \sim \pi(y|x)} [r(x, y)] - \beta E_{y \sim \pi(y|x)} \left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} \right] \right]$
 $= E_{x \sim \mathbb{D}} E_{y \sim \pi(y|x)} \left[r(x, y) - \beta \log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} \right]$

- 提取 $-\beta$: $\max_{\pi} E_{x \sim \mathbb{D}} E_{y \sim \pi(y|x)} \left[-\beta \left(-\frac{1}{\beta} r(x, y) + \log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} \right) \right]$

- 即 $\min_{\pi} E_{x \sim \mathbb{D}} E_{y \sim \pi(y|x)} \left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} - \frac{1}{\beta} r(x, y) \right]$

$$-\frac{1}{\beta} r(x, y) = \log \exp \left(-\frac{1}{\beta} r(x, y) \right)$$

$$= \log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} + \log \exp \left(-\frac{1}{\beta} r(x, y) \right) = \log \left(\frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} \cdot \exp \left(-\frac{1}{\beta} r(x, y) \right) \right)$$

- $\min_{\pi} E_{x \sim \mathbb{D}} E_{y \sim \pi(y|x)} \left[\log \left(\frac{\pi(y|x)}{\pi_{\text{ref}}(y|x) \exp \left(\frac{1}{\beta} r(x, y) \right)} \right) \right]$

$$\tilde{q}(y|x) := \pi_{\text{ref}}(y|x) \exp \left(\frac{1}{\beta} r(x, y) \right) \text{ 不再满足对 } y \text{ 求和为 } 1$$

不是一个概率分布

- $Z(x) := \sum_y \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right)$

不依赖于 y , 也不依赖于 π

$$q(y|x) := \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right) = \frac{\tilde{q}(y|x)}{Z(x)}$$

$$\log \frac{\pi(y|x)}{\tilde{q}(y|x)} = \log \frac{\pi(y|x)}{Z(x)q(y|x)} = \log \frac{\pi(y|x)}{q(y|x)} - \log Z(x)$$

我们的优化目标实际上变成了

$$\min_{\pi} D_{\text{KL}}(\pi(y|x) \| q(y|x))$$

最优解就是 $q(y|x)$

$$\pi^*(y|x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right)$$

终于!

我们无法算出 $\sum_y \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right)$

- 算不出来，那就想办法消掉 $Z(x)$

- $\pi^*(y|x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right)$

- $\log \pi^*(y|x) = \log\left(\frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right)\right)$
 $= \log \frac{1}{Z(x)} + \log \pi_{\text{ref}}(y|x) + \log \exp\left(\frac{1}{\beta} r(x, y)\right)$

$$\log \pi^*(y|x) = -\log Z(x) + \log \pi_{\text{ref}}(y|x) + \frac{1}{\beta} r(x, y)$$

$$\frac{1}{\beta} r(x, y) = \log \pi^*(y|x) - \log \pi_{\text{ref}}(y|x) + \log Z(x)$$

- $\frac{1}{\beta} r(x, y) = \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \log Z(x)$
- $r(x, y) = \beta \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x)$

$$r(x, y) \propto \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)}$$

从概率反推出奖励，这是符合直觉的

- $r(x, y_2) - r(x, y_1) = \beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)} - \beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)}$

- $p^*(y_1 \succ y_2 | x) = \frac{\exp(r(x, y_1))}{\exp(r(x, y_1)) + \exp(r(x, y_2))}$

$$\frac{e^A}{e^A + e^B} = \frac{1}{1 + e^{B-A}} = \sigma(A - B) \quad \sigma(t) = \frac{1}{1 + e^{-t}}$$

- $p^*(y_1 \succ y_2 | x) = \sigma(r(x, y_1) - r(x, y_2))$

继续推导

- $p^*(y_1 \succ y_2 \mid x) = \sigma \left(\beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)} - \beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)} \right)$

人类偏好变成了一个只依赖于策略相对参考策略的log概率比差

- 我们把最优策略 π^* 换成可训练的策略 π_θ

$$p_\theta(y_w \succ y_l \mid x) = \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right)$$

$$\max_{\theta} E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

$$\mathcal{L}_{\mathcal{DPO}}(\theta; \pi_{\text{ref}}) = -E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

DPO如何更新

- $\mathcal{L}_{\mathcal{DPO}}(\theta) = -\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right)$

$$u = \hat{r}_{\theta}(x, y_w) - \hat{r}_{\theta}(x, y_l) = \beta \left(\log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right)$$

- loss简化为了

$$\mathcal{L} = -\log \sigma(u)$$

- $\mathcal{L}$ 对 θ 求梯度:

链式法则

$$\nabla_{\theta} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial u} \cdot \nabla_{\theta} u$$

$$\nabla_{\theta} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial u} \cdot \nabla_{\theta} u$$

第一项: $\frac{\partial}{\partial u} (-\log \sigma(u)) = -\frac{1}{\sigma(u)} \cdot \sigma'(u)$

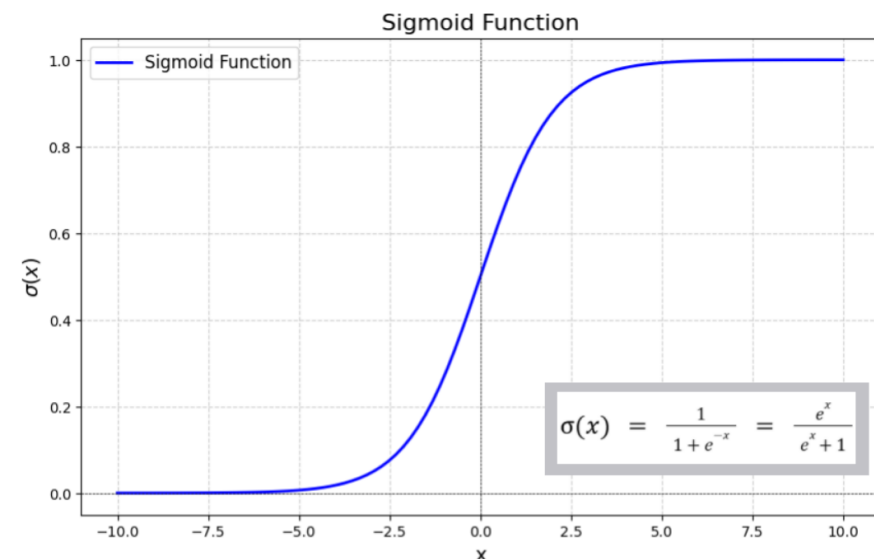
$$\sigma'(u) = \sigma(u)(1 - \sigma(u)), \quad 1 - \sigma(u) = \sigma(-u)$$

$$\frac{\partial}{\partial u} (-\log \sigma(u)) = -\sigma(-u)$$

第二项:

$$u = \beta [(\log \pi_{\theta}(y_w) - \log \pi_{\text{ref}}(y_w)) - (\log \pi_{\theta}(y_l) - \log \pi_{\text{ref}}(y_l))]$$

$$\nabla_{\theta} u = \beta (\nabla_{\theta} \log \pi_{\theta}(y_w | x) - \nabla_{\theta} \log \pi_{\theta}(y_l | x))$$



$$\begin{aligned} & \nabla_{\theta} \mathcal{L}_{\mathcal{DP}\mathcal{O}}(\pi_{\theta}; \pi_{\text{ref}}) \\ &= -\beta E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{当奖励估计错误时权重更高}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{增加 } y_w \text{ 的似然}} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{降低 } y_l \text{ 的似然}} \right] \right] \end{aligned}$$

监控量

- Reward Accuracy

$$Acc = E[\mathbf{1}(r^+ > r^-)]$$

- Reward Margin

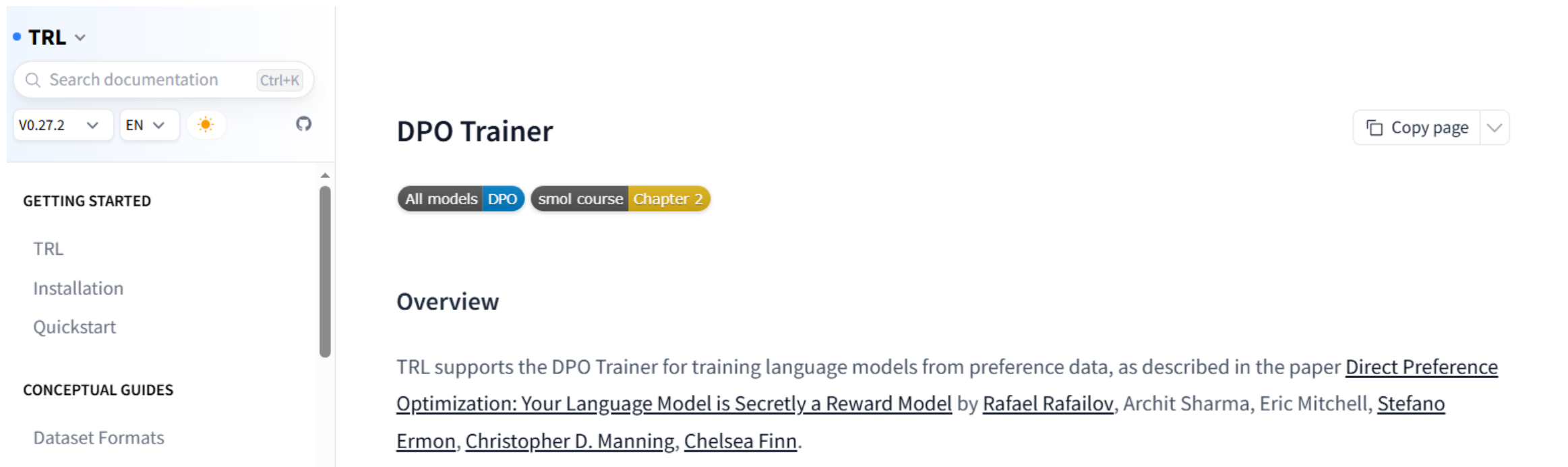
$$Margin = E[r^+ - r^-]$$

- Rewards Chosen & Rewards Rejected

$$\hat{r}_\theta(x, y_w) \quad \hat{r}_\theta(x, y_l)$$

快速上手

- 模型： Qwen/Qwen2-0.5B-Instruct
- 数据集： openbmb/UltraFeedback
- 教程： https://huggingface.co/docs/trl/dpo_trainer



The screenshot shows the Hugging Face documentation interface for the TRL (Transformer Reinforcement Learning) library. The left sidebar contains a navigation menu with sections like 'GETTING STARTED' (including TRL, Installation, Quickstart) and 'CONCEPTUAL GUIDES' (including Dataset Formats). The main content area is titled 'DPO Trainer' and includes a breadcrumb trail: 'All models > DPO > smol course > Chapter 2'. Below this is an 'Overview' section that states: 'TRL supports the DPO Trainer for training language models from preference data, as described in the paper [Direct Preference Optimization: Your Language Model is Secretly a Reward Model](#) by [Rafael Rafailov](#), Archit Sharma, Eric Mitchell, [Stefano Ermon](#), [Christopher D. Manning](#), [Chelsea Finn](#).'

TRL ▾

🔍 Search documentation Ctrl+K

V0.27.2 ▾ EN ▾ 🌞 🔄

GETTING STARTED

- TRL
- Installation
- Quickstart

CONCEPTUAL GUIDES

- Dataset Formats

DPO Trainer

All models DPO smol course Chapter 2

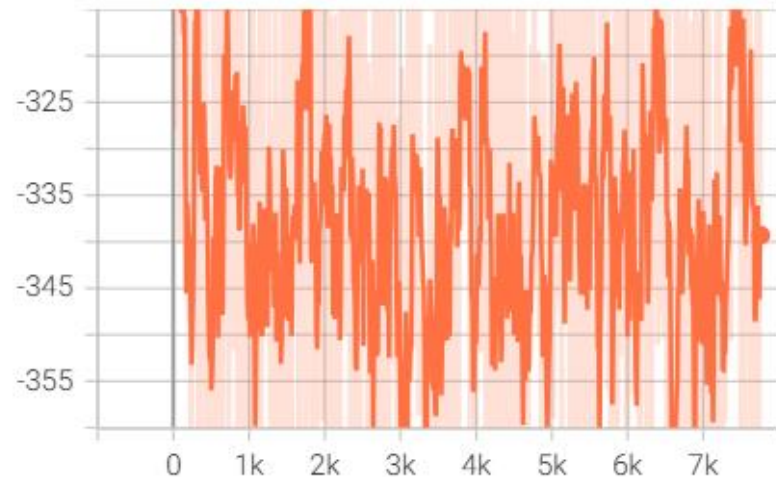
Overview

TRL supports the DPO Trainer for training language models from preference data, as described in the paper [Direct Preference Optimization: Your Language Model is Secretly a Reward Model](#) by [Rafael Rafailov](#), Archit Sharma, Eric Mitchell, [Stefano Ermon](#), [Christopher D. Manning](#), [Chelsea Finn](#).

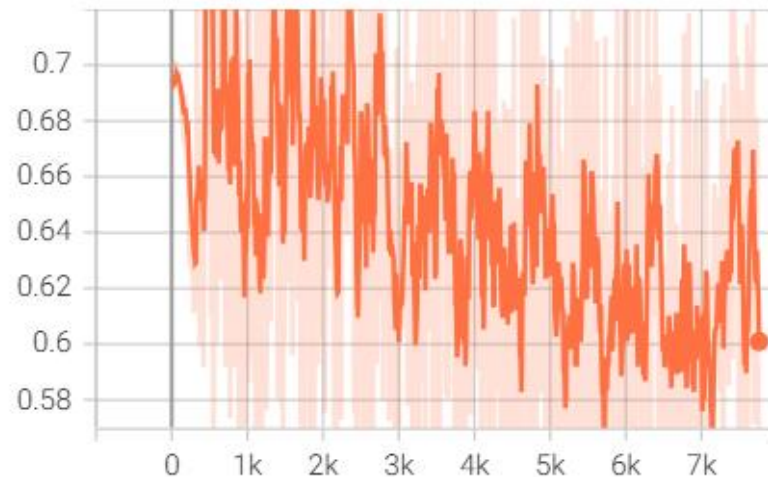
📄 Copy page ▾

<https://github.com/ZHAOops/dpo-qwen2-0.5b>

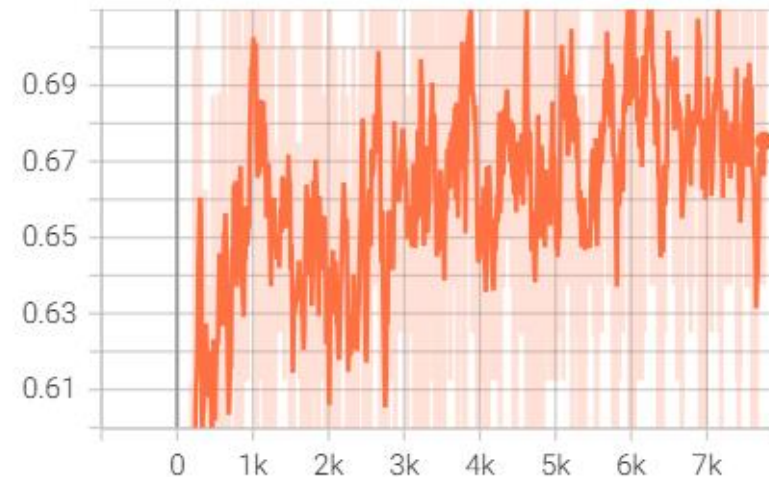
train/logps/rejected
tag: train/logps/rejected



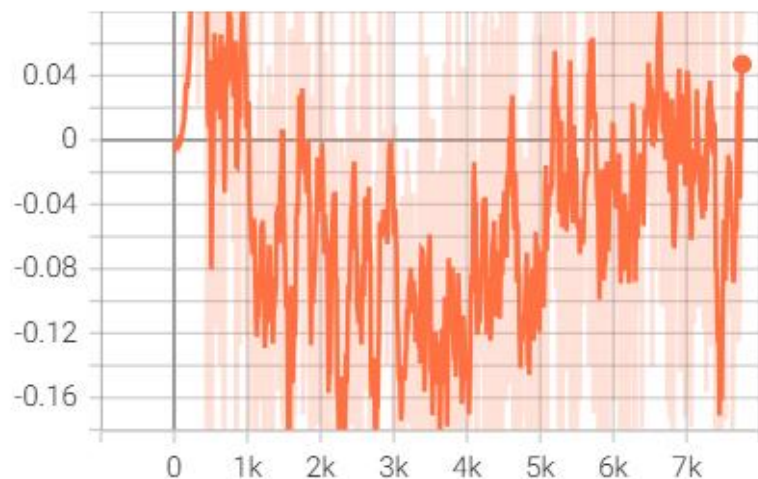
train/loss
tag: train/loss



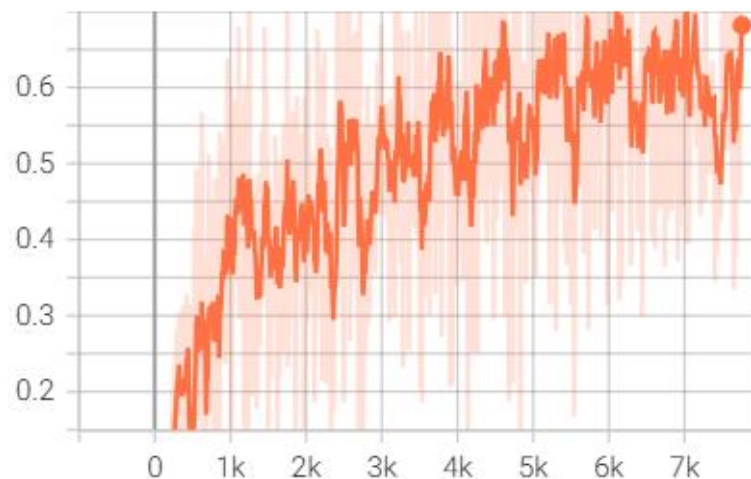
train/rewards/accuracies
tag: train/rewards/accuracies



train/rewards/chosen
tag: train/rewards/chosen



train/rewards/margins
tag: train/rewards/margins



train/rewards/rejected
tag: train/rewards/rejected

